

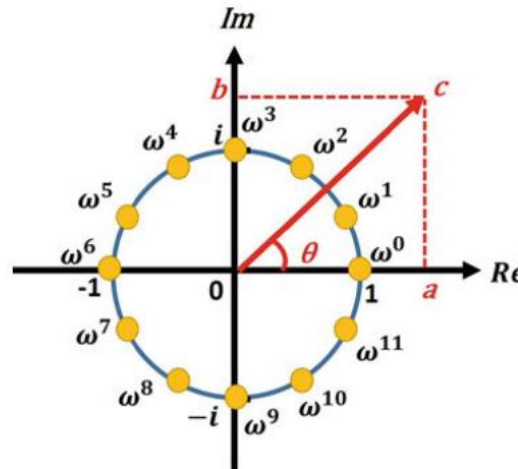
Chapter 24:

Quantum Fourier Transform I

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The N -th Root of Unity

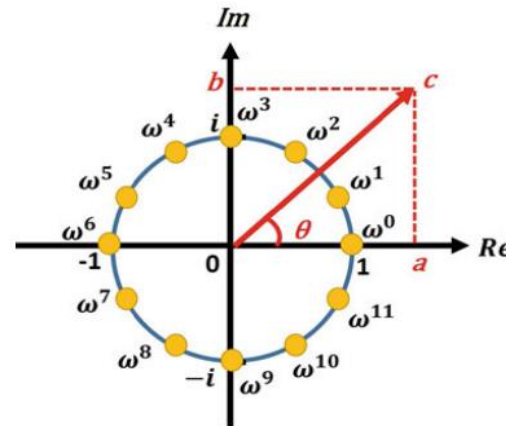
- Some of the important properties and identities of the complex numbers, particularly those related to the **N -th root of unity**. This is a very important warm-up to understand the **Quantum Fourier Transform (QFT)**.
- A complex number, c , has two components, namely the real and the imaginary parts. For example, $c = a + bi = 1.5 + 1.4i$, in which a and b are the real numbers and $a = 1.5$, $b = 1.4$, and $i = \sqrt{-1}$. This is equivalent to a 2D real space vector, and we can draw it on a 2D plane with the imaginary (real) part as the vertical (horizontal) axis as Pic. 33.



Pic. 33 Illustration of the complex plane and the N -th root of unity with $N = 12$.

The N -th Root of Unity

- Each point on the plane represents a complex number (or a vector pointing from the origin to that point).
- Each point is *NOT* a quantum state vector, and it is still just a number. It is used as the **coefficients** for the state vectors.
- We may also represent a complex number using the length of the vector, $|c|$, and its angle to the positive real axis, θ . That is $c = |c|e^{i\theta} = |c|(\cos \theta + i \sin \theta)$
- This is a **special set** of complex numbers with **lengths equal to 1**, and they lie on the unit circle in Pic. 33.

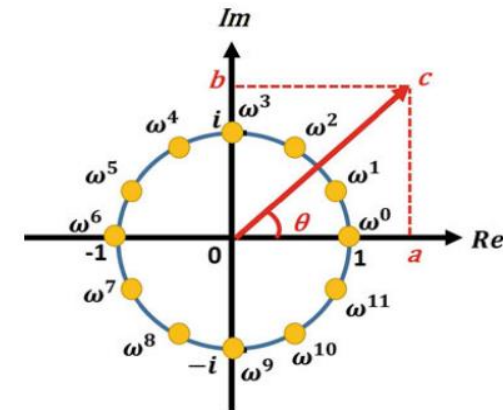


Pic. 33 Illustration of the complex plane and the N -th root of unity with $N = 12$.

The N -th Root of Unity

- They can be expressed simply as $e^{i\theta} = (\cos \theta + i \sin \theta)$ because $|c| = 1$.
- Among them, some of them (yellow dots), z , are even more special because they are the N -th roots of unity, which means $z^N = 1$.
- We can easily check that $z = e^{i2\pi m/N}$, with m being an integer, satisfies $z^N = 1$ because

$$z^N = (e^{i2\pi m/N})^N = e^{i2\pi m} = \cos 2\pi m + i \sin 2\pi m = 1$$
- The N -th roots of unity reside and spread evenly on the unit circle. We define $\omega = e^{i2\pi/N}$, and all can be expressed as ω^m , where m runs from 0 to $N-1$. For example, when $m = 0$, $\omega^0 = 1$. Pic. 33 shows the distribution of the N -th roots of unity on the unit circle when $N = 12$.



Pic. 33 Illustration of the complex plane and the N -th root of unity with $N = 12$. ¹⁵⁰

The N -th Root of Unity

- As m increases, ω^m moves counter-clockwise, and eventually, it repeats after one cycle (corresponds to increasing the exponent by N). This can be seen from proved using

$$\omega^{m+N} = e^{i2\pi(m+N)/N} = e^{i2\pi m/N} e^{i2\pi N/N} = e^{i2\pi m/N} = \omega^m$$

- The second property is that the sum of all N -th roots of unity for a given N is zero. This can be seen from the picture again. Since they are all evenly and symmetrically distributed, their corresponding vectors sum to null. This can also be proved as

$$\begin{aligned} \sum_{m=0}^{N-1} \omega^m &= \omega^0 + \omega^1 + \dots + \omega^{N-1} \\ &= \frac{\omega^0 - \omega^N}{\omega^0 - \omega} = \frac{1 - 1}{1 - \omega} = 0 \end{aligned}$$

Here we have used the summation equation of a geometric series with the first term being ω^0 , the ratio being ω , and the number of terms being N .

The N -th Root of Unity

- We also used that $\omega^N = \omega^{0+N} = \omega^0$. Similarly, for any integer $q \neq 0$,

$$\begin{aligned} \sum_{m=0}^{N-1} \omega^{mq} &= \omega^{0q} + \omega^{1q} + \dots + \omega^{(N-1)q} \\ &= \frac{\omega^{0q} - (\omega^q)^N}{\omega^{0q} - \omega^q} = \frac{1 - 1}{1 - \omega^{0q}} = 0 \end{aligned}$$

- For this geometric series, the first term is ω^{0q} , the ratio is ω^q , and the number of term is N . $(\omega^q)^N = 1$ is just because $(\omega^q)^N = (\omega^N)^q = 1^q = 1$. It represents the **destructive interference** we see in many quantum algorithms.
- However, if $q = 0$,

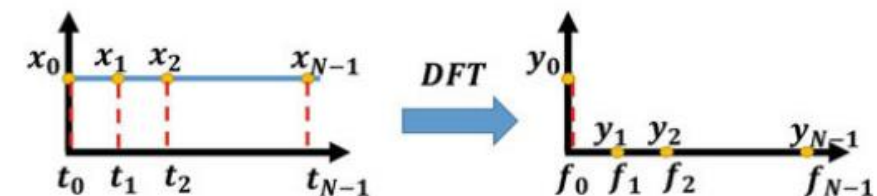
$$\sum_{m=0}^{N-1} \omega^{mq} = \sum_{m=0}^{N-1} \omega^0 = \sum_{m=0}^{N-1} 1 = N$$

- Because there are N 1's. This represents the **constructive interference** we see in many quantum algorithms. The final property is that $\omega^{N-m} = \omega^{-m}$. It can be proved as

$$\omega^{N-m} = e^{i2\pi(N-m)/N} = e^{i2\pi N/N} e^{i2\pi(-\frac{m}{N})} = e^{i2\pi(-\frac{m}{N})} = \omega^{-m}$$

Discrete Fourier Transform

- Discrete Fourier Transform (DFT) is the transformation (or mapping) of a discrete data series (perceived as in one domain e.g. time) into another data series (perceived as in another domain e.g. frequency).
- For example, if we sample the voltages of an electrical signal throughout the time, you will get a series of data (a series of voltages). By performing a DFT on the voltage series, you will get another series of data corresponding to the strength of the signal due to various frequency components. Both data series contain the same amount of information.
- Pic. 34 shows an example. The signal is constant (so it is a DC voltage). It is sampled at different times, from t_0 to t_{N-1} , and the sampled values form a series of numbers, $\{x_0, x_1, \dots, x_{N-1}\}$. After DFT, this series is transformed into another series of numbers $\{y_0, y_1, \dots, y_{N-1}\}$. In this particular case, each y_i represents the strength of various frequency components from f_0 to f_{N-1} .



Pic. 34 Illustration of Discrete Fourier Transform ¹⁵³

Discrete Fourier Transform

- Since it is a DC signal, only $f_0 = y_0\text{Hz}$ has a non-zero value as expected (meaning that it has no contribution due to any AC components). We can say that the t_i 's and the f_i 's form two sets of basis vectors with x_i and y_i being their “magnitudes,” respectively. They are just like the $\hat{x}$, $\hat{y}$, $\hat{z}$ in the 3D real space.
- In this application, they are perceived to be two different sets of basis (i.e. time vs. frequency). But mathematically, they can be the same set of basis vectors because, as you will see below, they (the basis vectors) do not appear in any equations except they mandate the locations of x_i and y_i .
- The exact content of the basis vector is not important to the operations after we have chosen the basis. So, we can make the series to be a vector and let them share the same basis.
- We then have $\vec{X} = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{pmatrix}$ and $\vec{Y} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N-1} \end{pmatrix}$. The DFT is defined as follow:

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega^{-kj} x_j = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-i2\pi kj/N} x_j$$

where $\omega = e^{i2\pi/N}$ is the N -th root of unity. The dimensions of the series/vectors are N . We see that the new series is just a linear combination of the old series weighted by the N -th roots of unity.

Discrete Fourier Transform

- This can also be represented in a matrix form as

$$\begin{aligned}\vec{Y} &= DFT\{\vec{X}\} \\ \vec{Y} &= \Omega \vec{X} \\ \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N-1} \end{pmatrix} &= \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.0} & \dots & \omega^{-0.(N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1).0} & \dots & \omega^{-(N-1).(N-1)} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{pmatrix}\end{aligned}$$

- We see that each column and row in the DFT matrix are composed of a power series of the N -th roots of unity. In particular, the component of the series after DFT (i.e. y_k) is the sum of a power series of the N -th roots of unity (i.e. ω^{-kj}) weighted by the components of the input series (i.e. x_j). If x_j is constant, then it is just purely a sum of a power series of the N -th roots of unity scaled by the x_j .

- E.g. Find $\vec{Y}$ if $\vec{X}$ is a normalized constant vector. $\vec{X} = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$ if it is normalized constant vector. Each component

has a magnitude of $\frac{1}{\sqrt{N}}$; therefore, its total length is $\sqrt{N(\frac{1}{\sqrt{N}})^2} = 1$. This means that $x_j = \frac{1}{\sqrt{N}}$.

- We can find y_k using $y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega^{-kj} x_j = \frac{1}{N} \sum_{j=0}^{N-1} \omega^{-kj}$. But we already know the answer. For $k = 0$, $y_0 = \frac{N}{N} = 1$. This is due to **constructive interference**. For $k \neq 0$, it becomes 0. This is due to **destructive interference**. Therefore, $\vec{Y} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, which is also a normalized vector.

Quantum Fourier Transform

- In DFT, we emphasized that the basis vectors of the input and output vectors (or series) can be the same.
- In engineering, we usually perceive them in different domains (e.g. time vs. frequency), but this is not necessary. If they are treated the same, **DFT is just a rotation of the vector on a given basis.**
- If we decide to use quantum basis states ($|j\rangle$) as the basis vectors, we can use DFT as usual. And this is called QFT.
- QFT is a quantum gate, and it needs to obey the properties of a quantum gate, namely, being unitary.
- In an N -dimensional space, we can assume $N = 2^n$ where n is the number of qubits, and the matrix of a QFT gate, U_{QFT} , is defined as

$$U_{QFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.0} & \dots & \omega^{-0.(N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1).0} & \dots & \omega^{-(N-1).(N-1)} \end{pmatrix} \dots \dots (24.1)$$

Note that U_{QFT} is a **symmetric matrix**. This is because $U_{QFT,ij} = U_{QFT,ji}$.

For example, $U_{QFT,01} = \omega^{-0.1} = \omega^{-1.0} = U_{QFT,10}$.

- Another equivalent way to define U_{QFT} is to define it by showing how it transforms the basis vectors.

Quantum Fourier Transform

- For basis vectors $|j\rangle$ and $|k\rangle$,

$$\mathbf{U}_{QFT}|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-kj} |k\rangle \dots \dots (24.2)$$

This can be checked easily using the fact that the k -th row and j -th column of $\mathbf{U}_{QFT}$ are given by $\langle k|\mathbf{U}_{QFT}|j\rangle$,

$$\mathbf{U}_{QFT,kj} = \langle k|\mathbf{U}_{QFT}|j\rangle = \left\langle k \left| \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-kj} |k\rangle \right| k' \right\rangle = \frac{1}{\sqrt{N}} \omega^{-kj}$$

- We can also check that Eqs. (24.1) and (24.2) are equivalent using numerical substitutions. Show that, for example, when $j = 1$, we have

$$\begin{aligned} \mathbf{U}_{QFT}|j\rangle &= \mathbf{U}_{QFT}|1\rangle \\ &= \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.0} & \dots & \omega^{-0.(N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1).0} & \dots & \omega^{-(N-1).(N-1)} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.1} \\ \omega^{-1.1} \\ \vdots \\ \omega^{-(N-1).1} \end{pmatrix} \\ &= \frac{1}{\sqrt{N}} (\omega^{-0.1}|0\rangle + \omega^{-1.1}|1\rangle + \dots + \omega^{-(N-1).1}|N-1\rangle) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-kj} |k\rangle \end{aligned}$$

Quantum Fourier Transform

- For $N = 2$, show how U_{QFT} transforms $|0\rangle$ and $|1\rangle$.

$$\begin{aligned}
 U_{QFT}|0\rangle &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-k0}|0\rangle \\
 &= \frac{1}{\sqrt{2}} (\omega^{-0.0}|0\rangle + \omega^{-1.0}|1\rangle) \\
 &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle
 \end{aligned}$$

$$\begin{aligned}
 U_{QFT}|1\rangle &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-k1}|0\rangle \\
 &= \frac{1}{\sqrt{2}} (\omega^{-0.1}|0\rangle + \omega^{-1.1}|1\rangle) \\
 &= \frac{1}{\sqrt{2}} (\omega^{-0.1}|0\rangle + e^{-i2\pi/2}|1\rangle) \\
 &= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle
 \end{aligned}$$

This is the same as the definition of the Hadamard gate! Therefore, a 1qubit U_{QFT} is just a Hadamard gate.

Quantum Fourier Transform

- Find the matrix representation of U_{QFT} when $N = 2$.
- This is a 1-qubit U_{QFT} , we have

$$\begin{aligned}
 U_{QFT} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \omega^{-0.0} & \omega^{-0.1} \\ \omega^{-(2-1).0} & \omega^{-(2-1).1} \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & \omega^{-1} \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & e^{-i2\pi/2} \end{pmatrix} \\
 &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}
 \end{aligned}$$

This is the matrix of the Hadamard gate.

- The first row and first column of any U_{QFT} are evaluated to be 1 because either the row or the column index is 0, and thus it is always $\omega^0 = 1$. For column j , the elements increase as a geometric series with ratio ω^{-j} (or decreases by ω^j). Therefore, we can rewrite as:

$$U_{QFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega^{-1} & \dots & \omega^{-(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-(N-1)} & \dots & \omega^{-(N-1)(N-1)} \end{pmatrix}$$

Quantum Fourier Transform

- Find the matrix representation of $\mathbf{U}_{QFT}$ when $N = 4$.
- This is a 2-qubit $\mathbf{U}_{QFT}$ gate. First, $\omega = e^{i2\pi/4} = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$. So, the element of the j -th column increases by $\omega^{-j} = i^{-j}$. Therefore,

$$\begin{aligned}\mathbf{U}_{QFT} &= \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i^{-1} & i^{-2} & i^{-3} \\ 1 & i^{-2} & i^{-4} & i^{-6} \\ 1 & i^{-3} & i^{-6} & i^{-9} \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}\end{aligned}$$

where elements in columns 0, 1, 2, and 3 increase by $i^{-0} = 1$, $i^{-1} = -i$, $i^{-2} = -1$, $i^{-3} = i$, respectively, from row to row.

Inverse Quantum Fourier Transform

- The inverse Quantum Fourier Transform (IQFT) is just the inverse matrix of QFT.
- It can be constructed by removing the negative sign in the exponents in the QFT matrix.

$$\begin{aligned}
 & \mathbf{U}_{IQFT} \\
 &= \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{0.0} & \omega^{0.1} & \dots & \omega^{0.(N-1)} \\ \omega^{1.0} & \omega^{1.1} & \dots & \omega^{1.(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ \omega^{(N-1).0} & \omega^{(N-1).1} & \dots & \omega^{(N-1).(N-1)} \end{pmatrix} \\
 &= \frac{1}{2} \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega^1 & \dots & \omega^{(N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{pmatrix}
 \end{aligned}$$

To prove it, we only need to show $\mathbf{U}_{IQFT}\mathbf{U}_{QFT} = \mathbf{U}_{QFT}\mathbf{U}_{IQFT} = \mathbf{I}$. We can see that the i -th row vector of $\mathbf{U}_{IQFT}$ is the complex conjugate of the i -th column vector of $\mathbf{U}_{QFT}$. By utilizing the definition of matrix multiplication and the unitary property, it means that $\mathbf{U}_{IQFT}$ and $\mathbf{U}_{QFT}$ are the inverse matrices of each other.

Chapter 25:

Quantum Fourier Transform II

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Another Definition of QFT and IQFT

- The QFT matrix is made exactly the same as the DFT matrix. This is a natural way to define the QFT matrix.
- Assume the vector is $|X\rangle = X_0|0\rangle + X_1|1\rangle + \dots + X_{n-1}|N-1\rangle$. This is a quantum state vector on the basis formed by the basis vectors $|0\rangle$ to $|N-1\rangle$. And assume after transformation, it becomes $|Y\rangle = X_0|0\rangle + X_1|1\rangle + \dots + X_{n-1}|N-1\rangle$. Then the transformation shares exactly the same equation as below because we are using the same matrix for DFT and QFT.

$$|Y\rangle = \mathbf{U}_{QFT}|X\rangle$$

$$\begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_{N-1} \end{pmatrix} = \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0.0} & \dots & \omega^{-0.(N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1).0} & \dots & \omega^{-(N-1).(N-1)} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{N-1} \end{pmatrix}$$

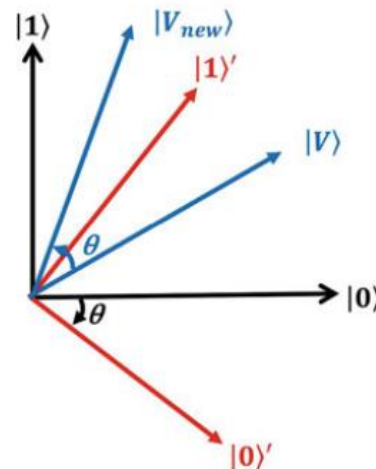
Therefore, each component of the transformed vector, $|Y\rangle$ still have the same equation as in the first part that is:

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \omega^{-kj} x_j$$

We call it QFT, because the matrix for transforming/rotating a quantum vector is the same as the DFT.

Another Definition of QFT and IQFT

- Pic. 33 shows an example of transforming a vector $|V\rangle$ by rotating it counter-clockwise by θ . The basis and basis vectors are not changed, and only the vector is rotated/transformed into $|V_{new}\rangle$ in the original $|0\rangle/|1\rangle$ basis.
- However, we can also perform a transformation of the basis in the reverse direction to achieve the same purpose.
- In Pic. 35, if we rotate the basis vector clockwise by θ , we achieve the same result. So, we need to rotate the basis using an inverse matrix of the one for rotating the vector. In the QFT case, applying QFT, U_{QFT} , to a vector is the same as applying IQFT, U_{IQFT} to the basis.



Pic. 35 Illustration of two equivalent transformations, transformation of basis (rotating basis vectors clockwise by θ) and transformation of vector (rotating vector by θ counter-clockwise)

Many-Qubit SWAP Gate

- The many-qubit SWAP gate is an important part of the QFT circuit.
- The definition of a 2-qubit swap gate is repeated here: $U_{SWAP}|ab\rangle = |ba\rangle$

An n -qubit SWAP gate is defined as $U_{SWAP,n}|X_0X_1X_2 \dots X_{n-3}X_{n-2}X_{n-1}\rangle = |X_{n-1}X_{n-2}X_{n-3} \dots X_2X_1X_0\rangle$

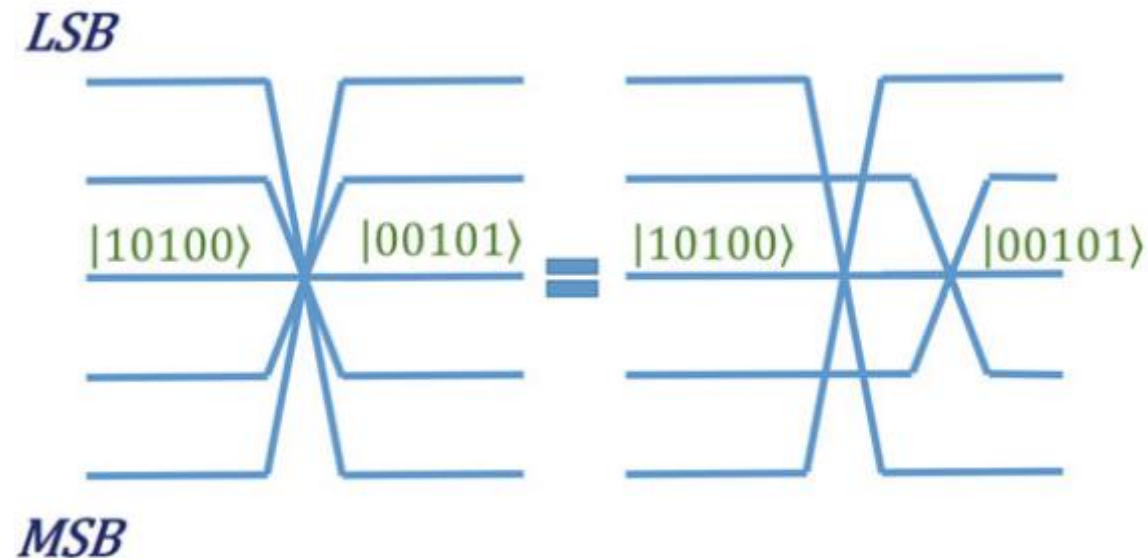
where $|X_0X_1X_2 \dots X_{n-3}X_{n-2}X_{n-1}\rangle$ is the basis vector of the 2^n -dimensional space, we can also write $|X_0X_1X_2 \dots X_{n-3}X_{n-2}X_{n-1}\rangle = |X_0\rangle|X_1\rangle|X_2\rangle \dots |X_{n-3}\rangle|X_{n-2}\rangle|X_{n-1}\rangle$. We see that we can implement the n -qubit SWAP gate using a series of 2-qubit swap gate.

- Let $U_{SWAP,i,j}$ be a 2-qubit SWAP gate swapping the i -th and j -th qubits. We have

$$\begin{aligned}
 & U_{SWAP,n}|X_0X_1X_2 \dots X_{n-3}X_{n-2}X_{n-1}\rangle \\
 &= U_{SWAP,\frac{n}{2}-1,n/2} \dots U_{SWAP,2,n-3} U_{SWAP,1,n-2} U_{SWAP,0,n-1} |X_0X_1X_2 \dots X_{n-3}X_{n-2}X_{n-1}\rangle \\
 &= U_{SWAP,\frac{n}{2}-1,n/2} \dots U_{SWAP,2,n-3} U_{SWAP,1,n-2} |X_{n-1}X_1X_2 \dots X_{n-3}X_{n-2}X_0\rangle \\
 &\quad \vdots \\
 &= |X_{n-1}X_{n-2}X_{n-3} \dots X_2X_1X_0\rangle
 \end{aligned}$$

Many-Qubit SWAP Gate

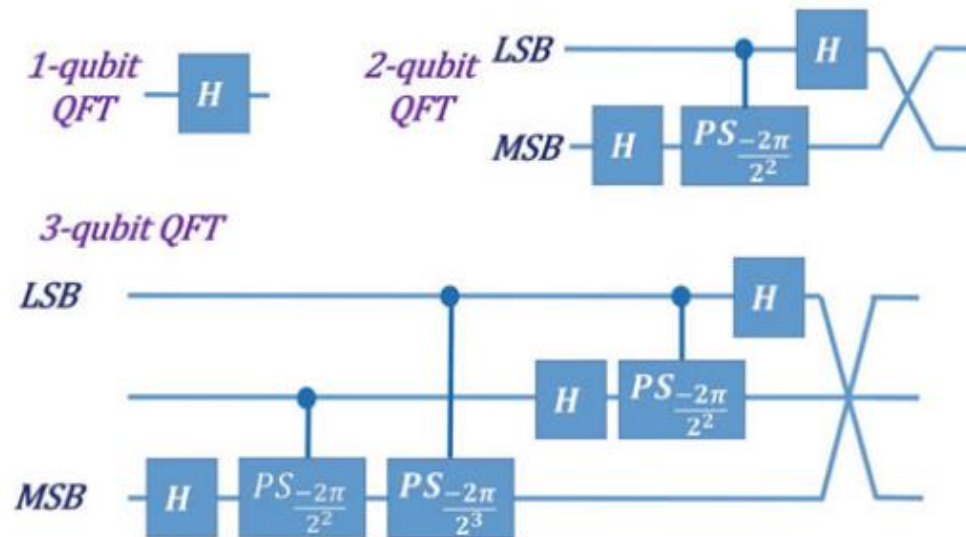
- Here we assume n is even. We first swap the 0-th and $(n - 1)$ -th qubits and then swap the 1-st and $(n - 2)$ -th qubits and continue until the $(n/(2 - 1))$ -th and $n/2$ -th qubits. If n is odd, the middle qubit need not to be swapped.
- Pic. 36 shows the implementation on IBM-Q. Note that the symbol of a 2-qubit SWAP gate is different from ours. It is used to test the $|10100\rangle$ basis vector. Therefore, two NOT gates are added to the 1st ($q4$) and the 3rd ($q2$) most significant qubits to generate $|10100\rangle$ from the initial state $|00000\rangle$. There are 5 classical bits to store the measured qubits. The MSB is at the bottom corresponding to bit 4.



Pic. 36 Quantum circuit for a 5-qubit SWAP gate (left). It can be implemented using two 2-qubit SWAP gates (right).

QFT Circuit

- Here we will describe how to construct a QFT circuit.
- A 1-qubit QFT circuit is just a Hadamard gate as we have proved.
- A 2-qubit QFT circuit is shown in Pic. 37. We need a 2-qubit SWAP gate, two Hadamard gates, and also a controlled phase shift gate.



Pic. 37 Quantum circuit of 1-, 2-, and 3-qubit QFT

- The matrix for a controlled phase shift gate is

$$U_{CPS,\phi} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix}$$

QFT Circuit

- From Pic. 37, we see that the phase needs to be shifted is $\phi = \frac{-2\pi}{2^2} = \frac{-\pi}{2}$. Therefore, the matrix is

$$U_{CPS, \frac{-2\pi}{2^2}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{-i\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -i \end{pmatrix}$$

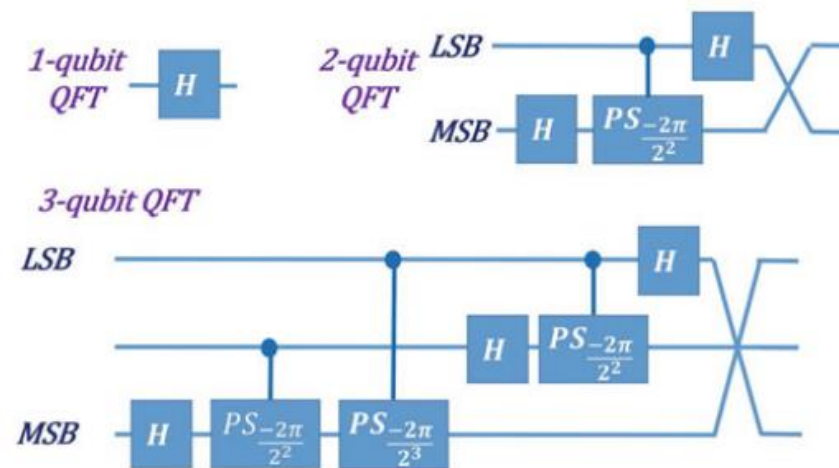
Therefore, the matrix of the 2-qubit QFT circuit is given by (note again that the vector goes from left to right in the circuit, but in matrix multiplication, the vector goes from right to left)

$$\begin{aligned} & U_{SWAP}(I \otimes H)U_{CPS, \frac{-2\pi}{2^2}}(H \otimes I) \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -i \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & -i & 0 & -i \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \\ 1 & i & -1 & -i \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix} \end{aligned}$$

Implementation of a 3-Qubit QFT Circuit

- Pic. 37, also shows the circuit for a 3-qubit QFT. We see that when we add one more qubit so that the total number of qubits is n , that qubit needs to go through a H gate, and then $(n-1)$ $U_{CPS,\phi}$ gates, with ϕ going from $\frac{-2\pi}{2^2}$ to $\frac{-2\pi}{2^n}$.
- For the 3-qubit QFT, we need a $U_{CPS,\frac{-2\pi}{2^3}}$ gate, which has a matrix of

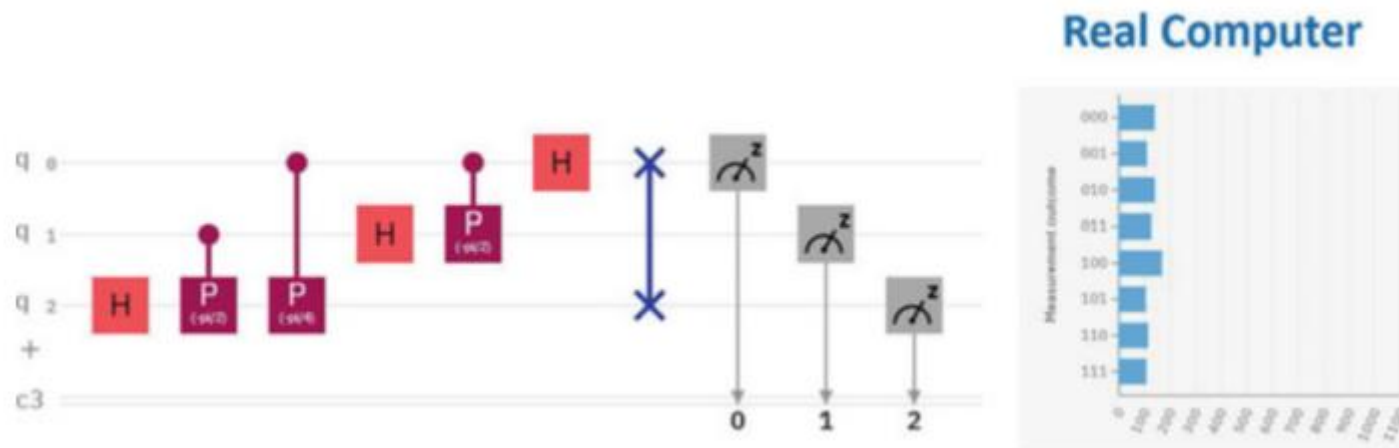
$$U_{CPS,\phi} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\frac{-2\pi}{2^3}} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\frac{-\pi}{4}} \end{pmatrix}$$



Pic. 37 Quantum circuit of 1-, 2-, and 3-qubit QFT

Implementation of a 3-Qubit QFT Circuit

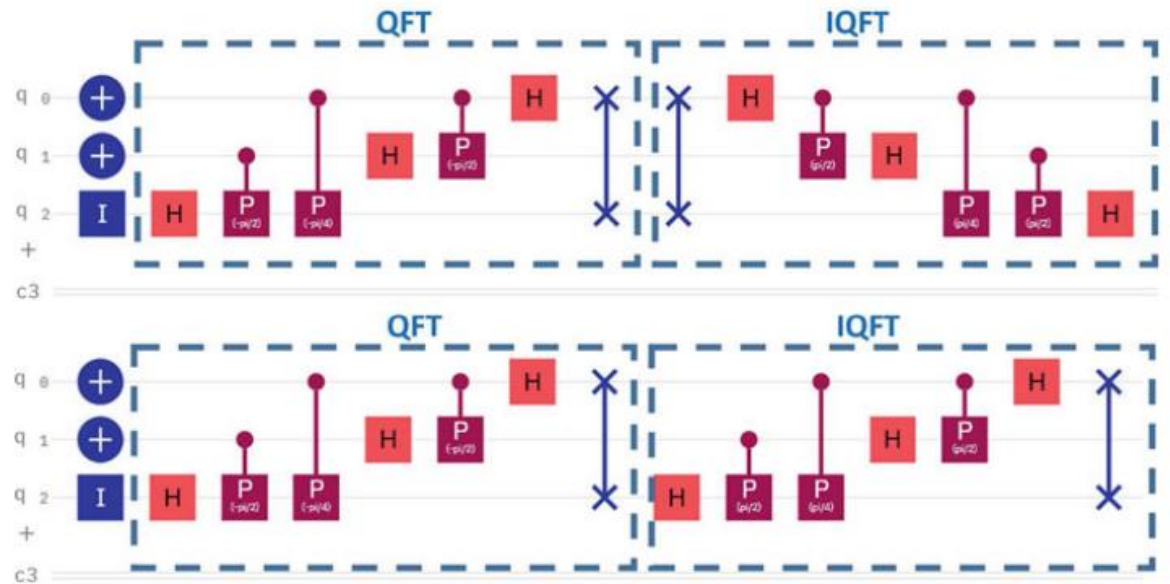
- Pic. 38, shows the implementation of the circuit for a 3-qubit QFT. Everything is a straightforward mapping to the circuit in Pic. 37. The only part that needs special attention is the $U_{CPS,\phi}$ gate.
- To create the $U_{CPS,\phi}$, one needs to choose the “Phase gate” icon first and then drop the “Control gate modifier” (the icon with a black dot) to it.
- The phase angles used from left to right are $-\pi/2$, $-\pi/4$, and $-\pi/2$ in this circuit. Since the input is $|000\rangle$ by default (the initialized quantum state), we expect the output to be an equal linear superposition of all basis vectors, and indeed the hardware execution shows the correction result besides with noise



Pic. 38 Quantum circuit of a 3-qubit QFT implemented on IBM-Q. The phase angles used from left to right are $-\pi/2$, $-\pi/4$, and $-\pi/2$. The hardware result is shown on the right

Implementation of IQFT

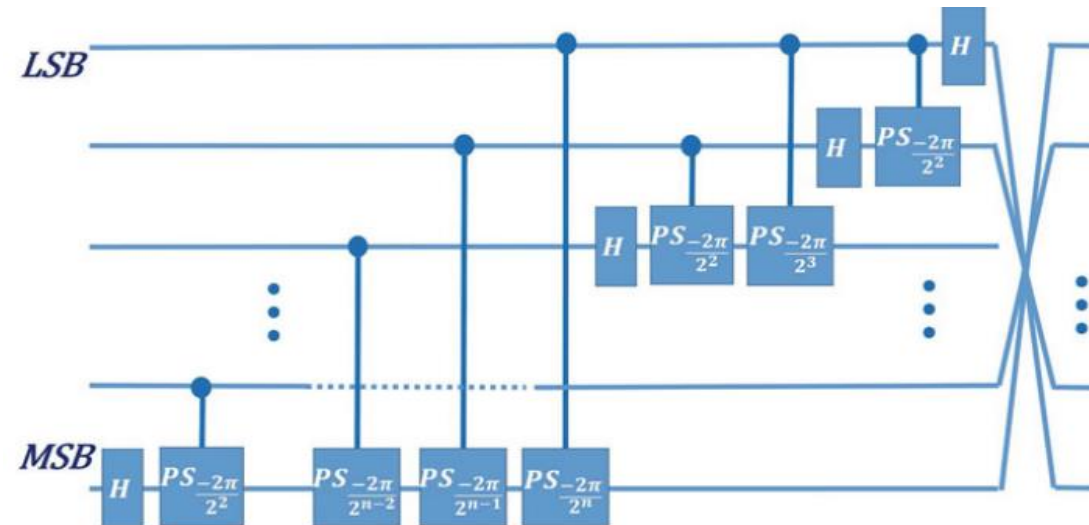
- By definition, IQFT is the inverse of QFT. Therefore, to construct the circuit for IQFT, we only need to reverse the arrangement of the gates in the QFT circuit and replace them with their inverse counterparts.
- Since the Hadamard gate and the SWAP gate are their own inverse, we can just leave them as they are.
- For the controlled phase shift gate, all we need to do is to change the angle to positive to get its inverse counterpart. So, we can just reverse the order of the QFT circuit and negate the phase angles to construct an IQFT circuit. This is shown at the top of Pic. 39.
- Another way to implement the circuit for IQFT is by realizing that IQFT and QFT have the same matrix form except the exponents have different signs. Phase shift term $e^{i\phi} = e^{-i2\pi/2^n}$ is just the ω^{-1} in the QFT for the 2^n dimension, and this is the only source for generating the ω in the final QFT or IQFT matrices. So, we can just use the QFT gates but negate all phases, i.e. change ϕ to $-\phi$ as shown at the bottom of Pic. 39.



Pic. 39 Implementation of IQFT after QFT. Top: IQFT is implemented by reversing the gates in QFT with ϕ changed to $-\phi$. Bottom: IQFT is implemented by using QFT circuits with ϕ changed to $-\phi$

General Circuit of QFT

- The general circuit for an n -qubit QFT is shown in Pic. 40.
- When we add one more qubit, that qubit needs to go through a H gate and then $n - 1$ $U_{CPS,\phi}$ gates, with ϕ going from $\frac{-2\pi}{2^2}$ to $\frac{-2\pi}{2^n}$. Therefore, for the MSB, it has $n - 1$ controlled phase shift gates after the Hadamard gate.
- To construct IQFT, as we have shown in the 3-qubit case, we can use the same circuit but use positive phases, or we can reverse the circuit and use positive phases. That is, changing $U_{CPS,\frac{-2\pi}{2^j}}$ to $U_{CPS,\frac{2\pi}{2^j}}$ for $j = 2$ to n .



Pic. 40 An n -qubit QFT circuit